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Extraterrestrial: The First Sign of Intelligent Life Beyond Earth

by Avi Loeb

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129 Highlights | 25 Notes

Highlight (Yellow) | Location 45

it has recently been surmised that forty-thousand-year-old cave paintings throughout Europe show that our distant ancestors tracked the stars. From poets to philosophers, theologians to scientists, we have found in the universe provocations for awe, action, and the advancement of civilization.

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the nascent field of astronomy, after all, that was the impetus for the scientific revolution of Nicolaus Copernicus, Galileo Galilei, and Isaac Newton that removed the Earth from the center of the physical universe. These scientists were not the first to advocate for a more self-deprecating view of our world, but unlike the philosophers and theologians who preceded them, they relied on a method of evidence-backed hypotheses that ever since has been the touchstone of human civilization's advancement.

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At the time of this writing, I serve as chair of Harvard University's Department of Astronomy, founding director of Harvard's Black Hole Initiative, director of the Institute for Theory and Computation within the Harvard-Smithsonian Center for Astrophysics, chair of the Breakthrough Starshot Initiative, chair of the Board on Physics and Astronomy of the National Academies, a member of the advisory board for the digital platform Einstein: Visualize the Impossible from the Hebrew University of Jerusalem, and a member of the President's Council of Advisors on Science and Technology in Washington, DC. It is my good fortune to work alongside many exceptionally talented scholars and students as we consider some of the universe's most profound questions.

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Throughout the expanse of space and over the lifetime of the universe, are there now or have there ever been other sentient civilizations that, like ours, explored the stars and left evidence of their efforts? I believe that in 2017, evidence passed through our solar system that supports the hypothesis that the answer to the last question is yes.

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Are we, both scientists and laypeople, ready? Is human civilization ready to confront what follows our accepting the plausible conclusion, arrived at through evidence-backed hypotheses, that terrestrial life isn't unique and perhaps not even particularly impressive?

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The world will end, of course, and most decidedly with a bang; our Sun, now about 4.6 billion years old, will in about 7 billion years turn into an expanding red giant and end all life on Earth. This is not up for debate, nor is it an ethical matter.

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What if an answer to "Are we alone?" presented itself, but it was subtle, fleeting, ambiguous? What if we needed to employ our powers of observation and deduction to their fullest extent in order to discern it?

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I consider the hypothesis that just such an answer was given to humanity on October 19, 2017.

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What if, for instance, that evidence was inert or defunct technology—the equivalent, perhaps, of a billion-year-old civilization's trash?

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as Galileo Galilei argued after looking through his telescope, evidence doesn't care about approval.

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Cosmic space and time have vast scales. There are more than a billion trillion sun-like stars in the observable volume of the universe, and even the luckiest among us live for merely 1 percent of a millionth of the lifetime of the Sun.

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should animate us to raise our ambitions, ask difficult questions that challenge our presumptions, and then set about rigorously pursuing evidence rather than likes.

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eleven days, starting on October 19, 2017. That was the length of time we had to observe the first known interstellar visitor.

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I submit that the simplest explanation for these peculiarities is that the object was created by an intelligent civilization not of this Earth. This is a hypothesis, of course—but it is a thoroughly scientific one. The conclusions we can draw from it, however, are not solely scientific, nor are the actions we might take in light of those conclusions.

Highlight (Yellow) and Note | Page 1

It intercepted the orbital plane, within which all of the planets in our solar system revolve around the Sun, on September 6, 2017. But the object's extreme hyperbolic trajectory guaranteed it would only visit, not stay. On September 9, 2017, the visitor reached its perihelion, the point at which its trajectory took it closest to the Sun. Thereafter, it began to exit the solar system; its speed far away—relative to our star, it was moving at about 58,900 miles per hour—more than ensured its escape from the Sun's gravity.

~16.3611 miles per second

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We assigned the object several official designations, finally landing on one: 1I/2017 U1. But our planet's scientific community and the public would come to know it simply as 'Oumuamua—a Hawaiian name reflecting the geographical location of the telescope used to discover the object.

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Among Hawaii's state-of-the-art telescopes are the ones that make up the Panoramic Survey Telescope and Rapid Response System (Pan-STARRS), a network of telescopes and high-definition cameras located at an observatory atop Haleakala, the dormant volcano that forms most of the island of Maui. One of the telescopes, Pan-STARRS1, has the highest-definition camera on the planet, and since it came online, the system overall has discovered most of the near-Earth comets and asteroids found in the solar system. But Pan-STARRS has another distinction—it gathered the data that initially tipped us off to 'Oumuamua's existence.

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moving too quickly to be bound by the Sun's gravity. This clue quickly led the astronomy community to agree that Weryk had found the first interstellar object ever detected in our solar system.

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The Hawaiian word 'oumuamua (pronounced "oh moo ah moo ah") is loosely translated as "scout." In its announcement of the object's official designation, the International Astronomical Union defined 'oumuamua slightly differently, as "a messenger from afar arriving first." Either way, the name clearly implies that the object was the first of others to come.

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It's not as if we didn't have a basis for comparison. Thousands of asteroids, dry rocks hurtling through space, are discovered every year, and the number of icy comets in our solar system is greater than our instruments can count. Interstellar visitors are far rarer than asteroids or comets. In fact, at the time of 'Oumuamua's discovery, we had never seen an object that originated outside of our solar system pass through it.

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in mid-November 2017, the International Astronomical Union—the organization that names newly identified objects in space—changed its designation for 'Oumuamua for the third and final time. Initially, the IAU had called it C/2017 U1; the C was for comet. Then it switched over to A/2017 U1; the A was for asteroid. Finally, the IAU declared it 1I/2017—the I stood for interstellar. By that point, that 'Oumuamua had come from interstellar space was one of the few things upon which everyone agreed. ...

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A scientist must go where the evidence leads, the old adage runs. There is humility in following the evidence, and it frees you from preconceptions that can cloud observations and insight.

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adulthood, a good definition of which might be “the point at which you have gathered enough experience that your models have a high success rate in forecasting reality.”

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Occam's razor

No more assumptions should be made than necessary. Minimalism

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I am awed by the order, by the fact that the laws of nature that we find here on Earth seem to apply out to the very edges of the universe. And for a long time, since well before the arrival of 'Oumuamua, I have harbored a corollary thought: the ubiquity of these natural laws suggests that if there is intelligent life anywhere else, it will almost certainly include beings who recognize these ubiquitous laws and who are eager to go where the evidence leads, excited to theorize, gather data, test the theory, refine, and retest. And eventually, just as humankind has done, to explore.

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Our civilization has sent five man-made objects into interstellar space: Voyager 1 and Voyager 2, Pioneer 10 and Pioneer 11, and New Horizons. This fact alone is suggestive of our unlimited potential to venture far out. So

too is the behavior of our more distant ancestors. For millennia, humans have journeyed to the farthest reaches of our planet seeking different lives or better lives or just seeking, often with a shocking level of uncertainty about what they would find or whether they would return.

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headline covering my presentation nicely captured the idea: “Be Humble, Earthlings.” The lecture was given a little less than a month before ‘Oumuamua—unknownst to Earthlings—passed the orbital plane of Mars, and I delivered it mere miles from Pan-STARRS1,

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... The first Pan-STARRS telescope, PS1, went online in 2008. Fifty years earlier, in 1958, another telescope had been built on the summit of Haleakala, but it was not used to study the stars; an animating fear at the time was Soviet satellites, and America wanted to be able to track them. Pan-STARRS, the Panoramic Survey Telescope and Rapid Response System, had a different objective: to detect comets and asteroids threatening to collide with Earth.

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a bygone Cold War helped set in motion an observatory of such complexity and technological richness that, decades later, in the cold, clear atmosphere atop a dead volcano, a sophisticated instrument in the array was able to detect ‘Oumuamua, which passed overhead just a few years after this particular telescope opened for business.

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About a decade ago, when my family first moved to Lexington, I discovered a broken branch on a young tree in the yard. I was advised by a local gardener to cut off the nearly severed limb. On closer inspection, I saw that living fibers were still linking it to the rest of the tree. I chose to bind the branch together with insulation tape. Today the branch rises to the sky far above my head, but the insulation tape remains at eye level. That tree sits near the house, visible from our windows. I point it out to my daughters in order to remind them that humble acts can have extraordinary consequences.

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Some of the most consequential decisions are made out of hopeful expectation of what might result.

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a lesson I eventually learned myself and that I have since tried to teach my own students—wasn’t about whether you should or shouldn’t follow the crowd but rather that you should take time to figure things out before acting. In deliberation, there is the humility of uncertainty. This, too, is an attitude toward life that I have worked to embrace, cultivate in my students at Harvard, and instill in my daughters. After all, it is what my parents sought to instill in me.

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Israelis then—as now—were aware that their nation’s continued survival depended on deliberating over the consequences of their choices.

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It is commonly thought that life is a collection of the places you visit. But this is an illusion. Life is a collection of events, and these are the results of choices, only some of which are ours to make.

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if humanity ever finds a habitable planet on which to establish an outpost of our civilization, the people who populate it will likely look and act much like the people of Beit Hanan. As human history shows, the immediate demands of settling outposts of a civilization recur. Of necessity, they would focus on the growing of food and the collective effort of supporting one another, from eldest to youngest. They would each need to be resourceful and multitalented, capable of repairing and engineering machinery, cultivating crops, and educating the young. I believe, too, that they would accommodate the life of the mind, even in their remoteness. And, I suspect that when their children came of age, they would be met with the same expectation I was: mandatory service to society.

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Talpiot mandated that we study physics and mathematics, which sounded close enough to philosophy, and studying anything at the university seemed far more exciting than slogging through the muck with a rifle on my back. Given the opportunity, I did my best to justify the government’s faith in me. It was also at this time that I started to realize that while philosophy asked the fundamental questions, it often couldn’t resolve them. Science, I was learning, might put me in a better position to pursue answers.

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Plasma is a state that matter reaches at high temperatures when atoms are broken into a sea of positively charged ions (atoms that have lost some electrons) and negatively charged free electrons.

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Playing to my strengths, the first major research frontier that I pioneered as an astrophysicist centered on when and how the atomic matter in the universe was transformed into a plasma. So began my fascination with the early universe, the so-called cosmic dawn, or the conditions under which the very stars were formed.

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Bahcall had entered college intending to study philosophy but quickly concluded that physics and astronomy provided a more direct route to the most basic truths of the universe.

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Astronomy, I was coming to understand, addresses questions that were previously restricted to the realms of philosophy and religion. Among these questions are the biggest of the big—“How did the universe begin?” and “What is the origin of life?” I have also found that staring out into the vastness of space, contemplating the start and end of everything, provides a framework for answering, “What is a life worth living?”

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I had learned that across all facets of life, taking the evidence presented to you and pursuing it with wonder, humility, and determination can change everything—if, that is, you are open to the possibilities contained in the data.

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Watching children step slowly into adulthood reminds me that the most mundane acts of our existence suggest something miraculous that can be traced back to the Big Bang.

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The success of Albert Einstein’s theory of relativity was not due to its formal elegance, which was developed over a series of publications from 1905 to 1915. It was not accepted until 1919, when Sir Arthur Eddington, secretary of the Royal Astronomical Society in England and an astronomer in his own right, confirmed the theory’s prediction that the Sun’s gravity would bend light. For scientists, what remains of a theory after its contact with data is what is deemed beautiful.

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Sometimes, by near accident, something exceptionally rare and special crosses your path. Life turns on your seeing clearly what’s in front of you.

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my experience taught me the value of freedom and diversity, specifically in the choice of research topics and the selection of collaborators, respectively.

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An honest survey of faculty across academia brings to mind men and women whose contributions are defined by opportunities extended and opportunities taken away. The same can be said of nearly all walks of life.

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I am deeply committed to helping young people fulfill their potential, even when that means challenging not just orthodox ideas but, sometimes, more pernicious orthodox practices. As part of this mission, I have worked hard to maintain—in my teaching and in my research—an approach to the world that some might consider childlike.

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In my experience, children follow their inner compasses more honestly and with fewer pretensions than many adults do. And the younger people are, the less likely they are to curb their thoughts to mirror the actions of others around them. This approach to science has opened me up to some of the more ambitious—some might say audacious—possibilities inherent in the topics I study.

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We still don't understand the nature of the main constituents of the universe, and so out of ignorance we label them dark matter (which contributes five times more to the cosmic-mass budget than the ordinary matter we are made of) and dark energy (which dominates both dark and ordinary matter and that causes, at least at present, the peculiar cosmic acceleration). We also do not understand what triggered the cosmic expansion or what happens inside black holes—two

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There is so much we do not know that I often wonder whether another civilization, one that had the benefit of pursuing science for a billion years, would even consider us intelligent. The possibility that they might extend us that courtesy, I suspect, will not be determined by what we know but by how we know it—namely, our fealty to the scientific method. It will be in our open-minded pursuit of data that confirms or disproves hypotheses that humanity's claim to any universal intelligence will stand or fall.

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Very often, what sets an astrophysicist's detective story in motion is the discovery of an anomaly in experimental or observational data, a piece of evidence that does not follow our expectations and that cannot be explained by what we know.

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first appreciate that in almost all scenarios, any two objects will rotate relative to each other when they pass. With that in mind, imagine a perfect sphere hurtling past the Sun as it makes its way through our solar system. The sunlight reflecting off its surface is unvarying, because the area of the tumbling sphere that faces the Sun is unvarying. Anything other than a sphere, however, will reflect the Sun's light by varying amounts as the object rotates. A football, for example, will reflect more light when one of its long sides faces the Sun and less light when, as it tumbles, its narrow sides face the Sun.

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The amount of reflected sunlight changed periodically by about a factor of ten (2.5 magnitudes) as 'Oumuamua rotated every eight hours. This implied that it had an extreme shape that was at least about five to ten times longer than it was wide when projected on the sky.

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This dramatic variability in its brightness told us that ‘Oumuamua’s shape was extreme, or at least five to ten times longer than it was wide.

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Its trajectory near the Sun meant that ‘Oumuamua should have had a very hot surface temperature, something that would have been visible to the infrared camera of the Spitzer Space Telescope, which NASA launched back in 2003. However, Spitzer’s camera was unable to detect any heat coming off ‘Oumuamua.

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Imagine setting down this book and taking a walk somewhere. You encounter other people. Perhaps they are strangers to you, and no doubt they all look different, but by their proportions, they are immediately recognizable as human. Among such passersby, ‘Oumuamua would be a person whose waist appears to be smaller than his or her wrist. Seeing such a person would cause you to question either your vision or your understanding of people. This was essentially the dilemma that astronomers faced as they began to interpret the early data about ‘Oumuamua. ...

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the object’s relatively small but extreme dimensions—with a length at least five to ten times greater than its width—allowed only two possible shapes. Our interstellar visitor was either elongated, like a cigar, or flat, like a pancake.

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If, as seems possible, ‘Oumuamua was a few times smaller than the upper limit of a few hundred yards that scientists presumed it to be, its reflectivity would approach unprecedented values—levels of brightness similar to a shiny metal.

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when ‘Oumuamua sped part of the way around the Sun, its trajectory deviated from what was expected based on the Sun’s gravitational force alone. There was no obvious explanation for why. This, for me, was the most eyebrow-raising bit of data we accumulated over the roughly two weeks we were able to observe ‘Oumuamua.

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With the publication in 1610 of his treatise *Sidereus Nuncius* (which translates to *Starry Messenger*), Galileo described his observations of the planets via a new telescope and declared—based on this evidence—his agreement with the heliocentric theory of the solar system. Galileo’s data implied that the Earth, along with all the other planets, revolved around the Sun. This ran directly counter to the teachings of the Catholic Church,

which accused Galileo of heresy. Following a trial during which it is claimed his accusers refused to even look through his telescope, Galileo was found guilty of heresy. He spent the rest of his life, nearly a decade, under house arrest.

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about ‘Oumuamua, testing our hypotheses against it, and, when others try to silence us, whispering to ourselves, “And yet it deviated.”

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Sir Isaac Newton’s first law of motion: “Every object persists in its state of rest or uniform motion in a straight line unless it is compelled to change that state by forces impressed on it.”

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A billiard ball sits on a pool table, unmoving, even as fourteen other balls careen around it; it remains unmoving until another ball strikes it. A solitary billiard ball sits on a pool table, unmoving, until a pool cue strikes it. A billiard ball sits, stationary, on a pool table until someone lifts up one end of the table. A billiard ball sits on a pool table, not moving, until a conical depression suddenly appears in the middle of the table. In either of the last two cases, gravity takes hold and the ball starts to move. Once moving, it will do so in a line dictated by the force that acted on it and will continue traveling along that line until another force acts on it.

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In June 2018, researchers reported that ‘Oumuamua’s trajectory deviated slightly, but to a highly statistically significant extent, from a path shaped by the Sun’s gravity alone. This is because it accelerated away from the Sun, being pushed by an additional force that declined roughly as the square of the distance from the Sun. What repulsive force, which opposes the attractive gravitational force, can be exerted by the Sun? Comets from the solar system show a deviation similar to ‘Oumuamua’s, but they are accompanied by a cometary tail of dust and water vapor from ice heated by sunlight.

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If the extra push that propelled ‘Oumuamua was from the rocket effect, as it is for comets, then our interstellar object should have lost a tenth of its mass in order for it to be propelled as much as it was. This is not a negligible amount of outgassing that could easily have been missed by our telescopes. But deep observations of the space around ‘Oumuamua did not reveal any trace of water, carbon-based gases, or dust, ruling out the possibility that it was being pushed by cometary vapor or visible dust particles. Moreover, it did not change its spin rate as it should have if one-sided jets were pushing it sideways, as they often do in comets. Then, too, such a massive evaporation would have changed the tumbling period of ‘Oumuamua, a phenomenon that is seen in solar system comets. No such change in the spin rate was recorded.

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We have observed many comets whose trajectories deviated from paths shaped by the Sun's gravity alone. We also know why that happens: in all cases, it is due to outgassing. But as I have just explained, 'Oumuamua showed no outgassing. And yet it deviated.

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IRAC was trained on 'Oumuamua for thirty hours as our interstellar visitor sped past the Sun. Had there been even trace carbon dioxide in 'Oumuamua's outgassing, the camera should have been able to observe it. But IRAC saw nothing—not a trail of gas behind the object and certainly not the object itself. (Interestingly, the Spitzer Space Telescope did not detect any heat being emitted from 'Oumuamua either, implying that it must be shinier than a typical comet or asteroid; that is the only way it could have reflected as much sunlight as it did while still being small enough not to produce much heat.)

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Just like Spitzer, SOHO and STEREO did not detect anything in the area; to these instruments, 'Oumuamua was invisible. This can only mean that 'Oumuamua had a "water production rate" that was "smaller than any of the previously reported limits by at least an order of magnitude."

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yet another difficulty with the outgassing-comet hypothesis, regardless of whether 'Oumuamua outgassed pure hydrogen or not. Its acceleration during deviation was smooth and steady.

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The odds of a naturally occurring comet composed of 100 percent hydrogen ice that outgasses from one location producing smooth acceleration? About the same as the odds of natural geological processes producing a space shuttle.

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Moreover, to account for 'Oumuamua's degree of deviation, a statistically significant portion of its total mass would have had to be outgassed. The nongravitational push was substantial enough—about 0.1 percent of the Sun's gravitational acceleration—that cometary outgassing could be responsible for the deviation only if the process had expended at least 10 percent of 'Oumuamua's mass.

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Then, too, the more material we have to imagine 'Oumuamua invisibly outgassed, the less likely it becomes we would have failed to observe it. And the smaller we have to imagine 'Oumuamua was in order to explain why we did not see the material it outgassed, the odder its luminosity and width-to-length ratio becomes—and the shinier it would have had to be.

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The object's smooth acceleration also defies the hypothesis that 'Oumuamua fractured around perihelion, breaking up and losing enough of its mass to explain its deviated path. Our instruments observed no debris indicating such a fracturing and disintegration; in fact, we saw evidence of the opposite: a smooth, steady acceleration.

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When I read reports that the extra force on 'Oumuamua declined inversely with the distance from the Sun squared, I wondered what could be pushing it if not outgassing or disintegration. The only explanation that came to mind was the sunlight bouncing off its surface like wind off a thin sail.

Right... Radiation disipates this way... Square if the distance. A solar sail...

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an astrophysicist at McMaster University, went back to the evidence to see if he could provide an answer. He evaluated all the brightness models the data allowed and concluded the likelihood of 'Oumuamua being cigar-shaped was small and the likelihood of 'Oumuamua being disk-shaped was about 91 percent. You should keep this percentage in mind when you see the umpteenth artist's rendering of 'Oumuamua as a cigar-shaped rock.

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Is there a simpler way to achieve the required surface-to-volume ratio for a pancake-shaped object? Yes, there is. You could build a thin, sturdy piece of equipment capable of deviating due to the effects of solar-radiation pressure to exactly such specifications.

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But this peculiar interest of mine became public only in 2007, when the cosmologist Matias Zaldarriaga and I proposed to eavesdrop on extraterrestrial radio signals.

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back in 1993 when I moved from Princeton's Institute for Advanced Study to Harvard. A question that preoccupied me then was: When had the stars first "turned on"—that is, when was the moment the laws of nature declared, "Let there be light"? Contemplating the birth of stars would lead me, years later, to ponder how civilizations might eavesdrop on one another.

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the effort to look far back in time to the earliest eons of the universe requires listening to the feeble radio emissions from primordial hydrogen, the most abundant element in the universe. This is best done with

telescopes capable of searching for that early hydrogen's signature, an intrinsic wavelength of twenty-one centimeters that is stretched (shifted toward redder—longer—wavelengths, what is called “redshifted”) to the scale of meters by the universe's expansion since the cosmic dawn. By the mid-2000s, this theoretically possible field of experimental research was becoming a reality. Long-wavelength radio telescopes were at last under construction;

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Our televisions, cell phones, computers, and radios all emit radiation at the very frequencies to which the MWA telescope was tuned in its effort to pick up radio emissions from primordial hydrogen in the early universe—just another example of how technological advances can hinder rather than help astronomers. All of this radio-wave pollution led me to a thought one day while I was eating my lunch alongside Matias and others. If our civilization emitted so much noise at that frequency, then perhaps other civilizations did too—extraterrestrial,

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it became something more serious to both of us when I learned of the Foundational Questions Institute (FQXi) inaugural request for outside-the-box projects. With no baggage of past activity on related topics and resting on our reputations as mainstream scientists, I suggested to Matias that we turn this intriguing anecdote into an original research project. The fact that we were cosmologists not associated with the Search for Extraterrestrial Intelligence (SETI) Institute, which has always stood outside of the circle of more fashionable scientific organizations and has less advanced radio detectors and analysis, gave the project more credibility, and funding.

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I have long been aware that within the discipline of astronomy, SETI faces hostility. And I have long found that hostility bizarre. Mainstream theoretical physicists now widely accept the study of extra-spatial dimensions beyond the three we are all familiar with—plainly put, height, width, and depth—and the fourth dimension, time. This is despite the fact that there is no evidence for any such extra dimensions. Similarly, a hypothetical multiverse—an infinite number of universes all existing simultaneously in which everything that could conceivably happen is happening—occupies many of our planet's most admired minds, again despite the fact that there is no evidence that such a thing is possible.

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The Milky Way hosts tens of billions of Earth-size planets with surface temperatures similar to our own. Overall, about a quarter of our galaxy's two hundred billion stars are orbited by planets that are habitable in the way Earth is, with surface conditions that allow liquid water and the chemistry of life as we know it. Given so many worlds—fifty billion in our own galaxy!—with similar life-friendly conditions, it's very likely that intelligent organisms have evolved elsewhere. And that's counting only habitable planets within the Milky Way. Adding all other galaxies in the observable volume of the universe increases the number of habitable planets to a zetta, or 10²¹—a figure greater than the number of grains of sand on all of the beaches on Earth.

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help direct attention and effort toward another question: Are we alone? In the years that followed my endeavor with Matias, I was drawn to more and more SETI-related issues. What would be an evidence-backed approach to answering its guiding question? And because it joined a list of topics that aroused my curiosity—the nature of black holes, the beginnings of the universe, the possibility of near-light-speed travel—I found myself keeping company with any scholar whose interests overlapped my own, including some scientists who were exclusively identified with the search for alien intelligence. Subsequently, Ed Turner, an astrophysicist at Princeton University, and I were the first to consider how one might go about seeking evidence of artificially produced light. We had the idea that we might try to find the glimmer of, say, a spacecraft or a city at great distances with our modern telescopes. Encouraged by Freeman Dyson, we then turned the question on its head and started to wonder whether you could see a city of the size and brightness of Tokyo on Pluto, which at that time was the most distant planet in our solar system. (It has since been reclassified as a dwarf planet.)

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It turns out that if you used an instrument of the technological sophistication of the Hubble Space Telescope and sought the signature of artificial light for long enough, you could indeed see Tokyo from the edge of the solar system. And you could tell that the light was intrinsic and not reflected sunlight based on how the source dimmed with increasing distance from the Sun.

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A billionaire entrepreneur from Silicon Valley, Yuri Milner radiates an intensity of purpose. He was born in the Soviet Union, studied theoretical physics at Moscow State University, received his MBA from the Wharton School of the University of Pennsylvania, and became a stunningly successful investor. Companies he has helped to support include Facebook, Twitter, WhatsApp, Airbnb, and Alibaba.

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By 2015, Yuri was thinking of other, more direct and ambitious ways of advancing scientific projects that excited him, so he launched the Breakthrough Initiatives. The focus was unambiguous. The project was to seek answers to two of the most fundamental questions confronting humanity: Are we alone? And can we, by thinking and acting together, make the great leap to the stars?

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Artist's illustration of Proxima b, the nearest habitable planet outside of our solar system. This planet, discovered in August 2016, has a mass of roughly one to two Earth masses and revolves around Proxima Centauri, a dwarf star with 12 percent the mass of the Sun, at a distance of 4.24 light-years from Earth. Proxima b has a surface temperature comparable to that of the Earth, but because of its proximity to its faint host star, it is believed to be tidally locked, with permanent day and night sides.

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Yuri had been fascinated by these questions ever since, at a young age, he read the book *Universe, Life, Intelligence*, published in 1962 by the Soviet astronomer Iosif Shklovskii. (Shklovskii later released in an English edition, *Intelligent Life in the Universe*, coauthored with the American astronomer Carl Sagan.) It may have helped that Yuri's parents named him after the famed Soviet cosmonaut Yuri Gagarin, who in 1961—the year of the younger Yuri's birth—became the first human to be launched into space.

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A chemical-propulsion rocket, which is what has sent all of our Earth-launched ships into space, would take approximately one hundred thousand years to reach Proxima b. Yuri was fifty-six, so given his stipulated timeframe—within his lifetime—a propulsion rocket was a nonstarter. To get to Proxima b within several decades, we needed a spacecraft capable of traveling at a fifth of the speed of light. Even if we used nuclear fuel, which has the highest energy density of all fuels (other than antimatter, which is unavailable), it would be impossible for a propulsion rocket to reach such speeds. And Newton's second law of motion—which states that the acceleration of an object depends on its mass and the force acting upon it—dictated that our spacecraft would have to weigh as little as possible too. To get an object to accelerate to the desired speed would take a tremendous amount of energy; the lighter the object, the less energy would be required. Correspondingly, our spacecraft's payload would have to be no more than a few grams. That led to another challenge. Not only would our spacecraft need to cover the vast distance in much less than one hundred thousand years, but once it reached Proxima Centauri, it had to be able to take pictures and send them back to Earth in a manner we could detect. It had to be light, small, and inexpensive to manufacture. This dictated that the camera and transmitter would be similar to what is in today's mobile phones. By our calculations, that technology, with some modifications, would suffice.

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We calculated that a 100-gigawatt laser beam capable of targeting a sail roughly the size of a person for a few minutes would propel the sail, along with its attached camera and communication device, to one-fifth of the speed of light by the time the spacecraft was five times as far away as the moon. This would be the spacecraft's open runway, so to speak.

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Expensive? Somewhat; it was on the order of one of the biggest science projects, like the Large Hadron Collider at CERN or the James Webb Space Telescope, but cheaper than the Apollo moonshot. (Many people who heard about the Starshot Initiative confessed that they hadn't been as excited about space exploration since the Apollo mission fifty years earlier.) But it would also be efficient. Once built, the launch system could be used to send lots of such craft—which are perhaps better thought of as probes, though we fell into the habit of referring to them as StarChips.

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Top: Artist's impression of Starshot, a project to push a lightsail from Earth with a powerful laser beam. Middle: An example of the lightweight electronic device (StarChip) that could be attached, along with a camera, to the sail. Bottom: Photograph of the solar LightSail 2 deployed by the Planetary Society on July 23, 2019, with the Sun showing through the thirty-two-square-meter sail.

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Two weeks later, I visited Milner's home in Palo Alto and announced that we had a plan that fulfilled the mandates he had set out. Within our lifetimes, it was technologically feasible to send a craft to Proxima Centauri. Yuri was pleased and excited, as was Pete.

Highlight (Yellow) and Note | Page 60

After several months of extensive discussions, they decided to publicly announce the Starshot Initiative at the observatory atop One World Trade Center in New York City on April 12, 2016. This was Yuri's Night, a commemoration of the launch of the first human into space, Yuri Gagarin, on April 12, 1961. There, on the stage with Yuri Milner, I was joined by Freeman Dyson and Stephen Hawking. The historic vision presented by the panel was recorded by TV crews and broadcast all over the world.

StarShot announce at WorldTrade1... starting@5:14-9:54 <https://www.youtube.com/watch?v=c7VPHGjT8A4> StarChip: gram scale wafer w/camera, photo thrusters, communication and nav equipment (microfabrication) LightSail: nanotech sail (nanotechnology) The laser wind... facilitated by advances in photonics (scaled to 100GigaWatt level). Over a few min... it can accelerate 100m mph or 20% of lightspeed. 1k x faster than today's spacecraft (1m x faster than car on highway) More here: <https://breakthroughinitiatives.org/initiative/3> 1st craft 'could' launch by 2036...

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Just seventeen months after that presentation, 'Oumuamua was discovered by the telescopes of Pan-STARRS.

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the first three of 'Oumuamua's identified anomalies—its unusual orbit without a tail, its extreme shape, and its luminosity—make

Highlight (Yellow) | Page 61

Those three traits—orbit, shape, and reflectivity—do not exhaust 'Oumuamua's weirdness, as we know.

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When I proposed that Bialy and I explain 'Oumuamua's deviation by way of sunlight, I was following the same scientific tenet I had always followed—a hypothesis that satisfied all the data ought to be considered. Bialy

checked the numbers and his excitement grew; the idea I had proposed looked like a viable possibility. This led to a new question: What would we need to assume about ‘Oumuamua’s size and composition to explain its deviation? The key question was how thin ‘Oumuamua had to be in order to possess the extreme area-to-volume ratio that accounted for its excess acceleration. We determined that ‘Oumuamua needed to be less than a millimeter thick for the force of sunlight to be effective.

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must have been designed, built, and launched by an extraterrestrial intelligence.

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The lightsail inference may seem outlandish, but getting to it did not require any wild leaps. Shmuel and I went down a logical path. We followed the evidence, and, in the grand tradition of the detective work of science, we hewed closely to a maxim of Sherlock Holmes: “When you have eliminated the impossible, whatever remains, however improbable, must be the truth.” Hence our hypothesis: ‘Oumuamua was artificial. We laid out these ideas in a paper titled “Could Solar Radiation Pressure Explain ‘Oumuamua’s Peculiar Acceleration?”

The paper ‘Could Solar Radiation Pressure Explain Oumuamua’s Peculiar Acceleration? <https://arxiv.org/pdf/1810.11490.pdf>

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“A quarter of all stars host a planet the size and surface temperature of the Earth,” I said into the camera. “It would be arrogant to think we are alone.”

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Just a year earlier, following reports of an anomaly having to do with hydrogen atoms (which had been found to be colder than expected in the early universe), I published a paper with another Harvard postdoc, Julian Munoz, showing that if the dark matter was made of particles that possessed a tiny electric charge, they would cool the cosmic hydrogen and account for the reported anomaly.

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What questions arise and what new avenues for pursuing answers suggest themselves when we entertain possibilities that are counter to our governing presumptions but align with the data we have?

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‘Oumuamua’s data carefully and found that only highly exotic explanations could account for the object’s anomalous behavior. To explain all the known facts, they were forced to imagine ‘Oumuamua was a fluffy object composed of material a hundred times more rarefied than air or that it was a comet composed of solid hydrogen ice. Scientists had to offer up “never seen before” options to explain Oumuamua’s proven peculiarities.

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The questions that the lightsail hypothesis raises, moreover, are intriguing. If we suppose that ‘Oumuamua is an exceedingly rare comet composed of frozen pure hydrogen, most of our questions dead-end. The same is true if we imagine it as a fluffy cloud of dust with sufficient internal integrity to hold together but still lightweight enough to be pushed by sunlight.

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I was accused of seeing lightsails wherever I looked. After all, my involvement in the Starshot Initiative had been announced just two years earlier and our stated goal was to send electronic chips to the nearest star by harnessing the power of lightsail technology. The interviewer for the German newspaper Der Spiegel put it with admirable bluntness: “According to a proverb, whoever has only a hammer will see nothing but nails.”

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One of my favorite activities is walking along the beach searching for seashells worth collecting. It is a pastime I indulge in while I’m on vacation, when I get to pick a beloved stretch of sand to stroll and study and I have the free time to do it.

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The more there is of something, the more likely it is you will encounter it.

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Let’s suppose that, rather than a plastic bottle, ‘Oumuamua was a seashell. An exotic seashell, certainly, but still a naturally occurring seashell. This line of reasoning has attracted the vast majority of scientists

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In 2018, a small group of scientists revisited the question of the abundance of ‘Oumuamua-like rocks in interstellar space. While studying the ability of Pan-STARRS to detect objects similar to ‘Oumuamua, they reached some general conclusions. Among them was the broadly agreed-upon insight that “many aspects of ‘Oumuamua are both intriguing and troubling.” But they also found that the number per unit volume of interstellar material necessary to have ‘Oumuamua be a random rock requires “mass ejection rates” that far exceed expectations, up to a quadrillion (10¹⁵)

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The rough analogy of seashells helps; you need a great many seashells in the sea to make discovering an intact one on a beach probable. The same applies to a naturally occurring ‘Oumuamua arriving in our solar system. For it to be a randomly encountered object, we need a lot of such objects in the universe, and to get at that density, we need objects released from both planet formation and Oort clouds. Of course, we have already established

that ‘Oumuamua wasn’t icy. (No outgassing, no ice.) So a naturally occurring ‘Oumuamua is very unlikely to have come from an Oort cloud. In short, if ‘Oumuamua was a natural object, it had to have been generated by planet formation. It also has to belong to an unknown class of objects generated by planet formation

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When I met my wife and realized how special she was, I married her. The many people I have encountered since then do not take away from her unique qualities; they only add to my sense of wonder at how rare she is.

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All the stars in the vicinity of the Sun are moving relative to one another. The average of their motions is called the local standard of rest. Amid the motion of all these stars, an object at the local standard of rest, or LSR, is comparatively still. And it is a comparatively rare occurrence.

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Sky path of ‘Oumuamua as seen from Earth, with the stages of the object’s progress labeled by date. The relative size of each circle gives a schematic sense of the changing distance of ‘Oumuamua along its apparent trajectory. Also shown is the direction of motion of the Sun in the local standard of rest, or LSR (the direction just left of the label “solar apex”). The fact that the object starts from that direction and gets close to us implies that it was initially in the LSR. Between September 2 and October 22, 2017, ‘Oumuamua’s trajectory moved from the local standard of rest to south of the ecliptic plane of the solar system (marked by the thin line).

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‘Oumuamua was occupying the LSR. Or at least, it was before it accelerated.

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Our own solar system’s trillion-comet Oort cloud is a case in point. Its icy shell is 100,000 astronomical units (AUs) distant from the Sun (each AU is the distance from the Earth to the Sun, or roughly 93 million miles). The Sun’s gravitational hold on the material making up the Oort cloud is far weaker than its hold on, say, the Earth. Way out there, a mild nudge of less than 2,200 miles per hour—what an encounter with a passing star might provide, for example—could be enough to send an object on its interstellar way.

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The hypothesis that intelligent extraterrestrials designed ‘Oumuamua to be at LSR raises the obvious question: Why would they bother? I can imagine any number of reasons. Perhaps they wanted to set up the interstellar equivalent of a stop sign. Or maybe it was more like a lighthouse—or, more simply, a signpost or navigation marker. A vast network of such buoys could act as a communication grid. Or it could be used as a trip wire, an alert system triggered when one of them was knocked out of LSR. In that spirit, perhaps its creators wanted to disguise its—and their—spatial origins. Putting an object at LSR effectively camouflages who put it there. Why?

Because math and a little knowledge of an object's trajectory is sufficient to trace that object's origins back to a launchpad; doing that is one of the primary purposes of the North American Aerospace Defense Command (NORAD). Consider as well that any intelligence with a grasp of math and a good map of the universe could trace back to Earth any of the interstellar ships we've launched from our planet's surface. That all of these analogies are terrestrial isn't just a reflection of the fact that so is this book's author. That human civilization has built buoys, grids of communication satellites, and early-warning detection systems tells us that it is probable other civilizations might do the same thing. What is more, these conjectures are plausible for the simple reason that any one of them is an outcome that humanity could engineer, build, and launch if we wished to. Our reasons needn't even be interstellar.

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A buoy. A grid of pods for communication. Signposts that an extraterrestrial civilization could navigate by. Launch bases for probes. Other intelligent living organisms' defunct technology or discarded technological trash. These all are plausible explanations for the 'Oumuamua mystery—plausible because here on Earth, humanity is already doing these things, albeit on a far more limited scale, and we would certainly consider replicating them if and when we explore out into interstellar space. What renders these hypotheses implausible is the inability to conjecture an extraterrestrial intelligence. Foreclose that possibility, and you moot all such explanations. Refuse to look through the telescope and it matters little if it does or doesn't show any compelling evidence.

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'Oumuamua was a luminous, thin disk at the LSR, and when it encountered the gravitational pull of the Sun, it deviated from a trajectory explicable by gravity alone, and it did so without visible outgassing or disintegration. These data points can be summed up as follows: 'Oumuamua was statistically a wild outlier. Using very conservative probabilities, based on its shape, rotation, and luminosity alone, a cometary 'Oumuamua would be a one-in-a-million naturally occurring object.

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remember, that 'Oumuamua's spin rate didn't change. Maybe just one in every thousand comets keeps a steady spin despite the significant mass loss implied by 'Oumuamua's nongravitational acceleration. If 'Oumuamua was one of those rare comets, we're now talking about a one-in-a-billion object.

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They were what led me to suggest to Shmuel Bialy that we search for another, more plausible hypothesis. And we could come up with just one that made sense for the nongravitational acceleration: 'Oumuamua's weirdly steady push was provided by sunlight. This was fully consistent with an important clue. The excess force acting on 'Oumuamua that caused it to deviate, observers noted, seemingly declined inversely with the square of its distance from the Sun, as one would expect if that force was provided by reflected sunlight. But solar radiation pressure isn't very powerful. If it were indeed responsible, we calculated, then 'Oumuamua had to be less than a millimeter thick and at least twenty meters wide. (The diameter depends on the object's reflectivity, which is

unknown. If 'Oumuamua were a perfect reflector, bouncing back 100 percent of the sunlight that hit it, in this super-thin scenario it would be twenty meters across.)

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Are we alone? This question is among the most fundamental humanity confronts. The moment we have a conclusive answer, negative or positive, is the moment we face profound realizations.

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there are few cosmological questions of equal importance. To be sure, it would be transformative to learn what preceded the Big Bang, where the matter sucked into a black hole goes, or what theoretical insights finally square relativity with quantum physics.

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The paper was an immediate sensation, heralding the birth of SETI and establishing the rationale for all subsequent searches for extraterrestrial intelligence with its concluding sentence: "The probability of success is difficult to estimate; but if we never search, the chance of success is zero." For me, it echoes a much older thought, one attributed to Heraclitus of Ephesus: "If you do not expect the unexpected, you will not find it." Cocconi and Morrison's paper also brings to mind the old saw about people with nothing but hammers seeing only nails everywhere they look.

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Cocconi and Morrison's paper immediately inspired the astrophysicist Frank Drake, also at Cornell. In 1960 he decided to conduct just the kind of search

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on the updraft of that interest, in early November 1961, Drake participated in an informal conference sponsored by the National Academy of Sciences at the National Radio Astronomy Observatory. It was there that he first articulated the Drake equation, which he used to estimate the number of actively communicating extraterrestrial civilizations.

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$N = R^* \times fp \times ne \times fl \times fi \times fc \times L$, where the terms are defined as: N: the number of species in our galaxy that possess the technology necessary for interstellar communication; R*: the rate of star formation in our galaxy; fp: the fraction of stars with planetary systems; ne: the number of planets in each system with environmental conditions amenable to life; fl: the fraction of planets on which life arises; fi: the fraction of planets on which intelligent life arises; fc: the fraction of intelligent life that develops sufficiently sophisticated technology to take part in interstellar communication; L: the duration of time such intelligent life is able to produce detectable signals.

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In 1992, in keeping with the search for N, the U.S. government gave 12.25 million dollars to NASA to initiate a radio astronomy program. The very next year, SETI funding was ended. At the time Congress withdrew its support and the funding, one senator, Richard Bryan of Nevada, declared, “Millions have been spent and we have yet to bag a single little green fellow.” There are few more succinct statements of the ignorance and flawed assumptions that have hampered humanity’s pursuit of the answer to “Are we alone?” The sum spent was paltry, comparatively, and the bar set for evidence of success was absurd. That said, the early SETI researchers rarely helped their own cause. Their near-exclusive focus on seeking radio and optical signals has set unhelpful presumptions, scientific and popular, as to what such exploration looks like and what sorts of projects merit funding. Only recently have we witnessed growing interest in seeking biosignatures, such as oxygen and methane in the atmosphere and large-scale algal blooms in distant oceans, and technosignatures, such as markers of industrial pollutants in planetary atmospheres and localized heat islands that suggest urban settlements.

Note | Page 95

Hmmmm.

Note | Page 97

Or supersymetry

Note | Page 98

Not exactly what the revered scientists and equations demonstrate in the world of physics i gather... thus the precedent and example others follow (einstein. Schrodinger)

Note | Page 102

Could write to NdGT...

Note | Page 109

Sherlock

Note | Page 133

If we found these a few 100m light years away, we might anticipate that by now via evolution...

Note | Page 133

BioSigs v TechnoSigs... existing archeologists v futurist astro_archeologists in search of alternate civilizations across much broader timescales

Note | Page 134

At odds with Dysons belief that surface area and mobility suit spread and discovery of intelligent life wrt whete our odds are higher of finding it throughout the universe

Note | Page 135

Abiogenesis

Note | Page 135

Did abiogenesis happen multiple times... Or did evolution take over after a single or some minimal number of instances

Note | Page 138

So similarly... He's Suggesting if we trained tech dev and usage on identifying other solar sails we might just find them and ET life evidence in so doing

Note | Page 141

Havent we i.mlicitly done this and not found such debris on earth... Hmph

Note | Page 141

This doesnt make sense... See prior comment/note. Hes going astray...

Note | Page 148

Explanation of phenomena by purposes served rather than by causes

Note | Page 156

And both would contain evidence of ethics and atrocity in the name of their belief or lack thereof

Note | Page 158

Alfie

Note | Page 162

intergalactic nonterrestrial law...

Note | Page 164

and is this how life arose on earth...

Note | Page 181

Have to be willing to try (some crazy stuff, that before doing may in some cases seem impossible)

Note | Page 181

Occam's razor: the simplest explanation may just be the right one
